

Sensory Robotic Arm

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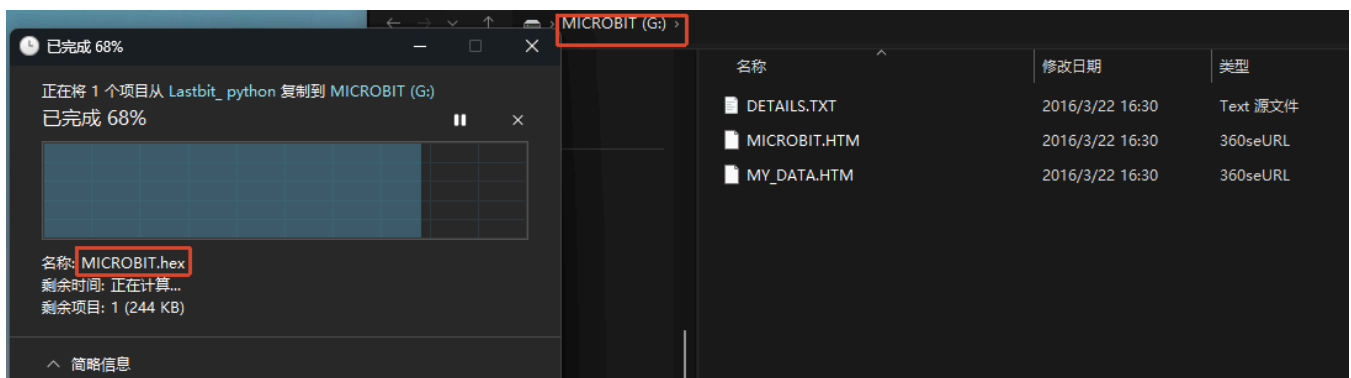
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Function Description

In this section, we will learn to use the accelerometer sensor to implement motion-sensing control of the robotic arm. By tilting the micro:bit, you can control the pitch and roll angles of the robotic arm for motion-sensing operation.

Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.

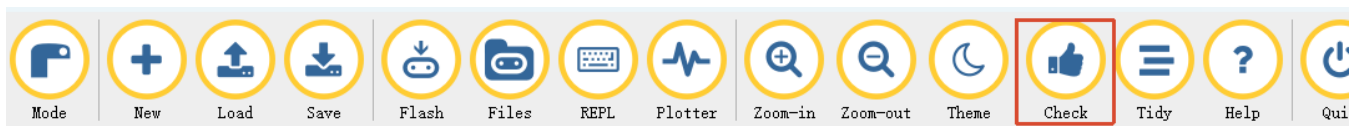
Click the `New` button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

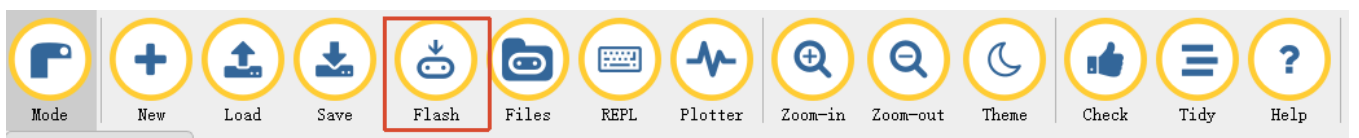
4. Now you can click `Check` to verify whether there are any problems with the code format.



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows `Copied code onto micro:bit.` at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

```
# 10. Motion Sensing Robotic Arm
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Servo angle
```

```

Servo_angle = 90
# Z-axis servo angle
servo_angle_z = 150
# Pitch proportional coefficient
KP = 1.2
# Roll proportional coefficient
KP_row = 0.8

# Initialize servos
Lastbit.set_all_servos(90)
Lastbit.set_servo('S2', servo_angle_z)
sleep(1000)

while True:
    # Get pitch angle
    angle_now = Lastbit.get_pitch()

    # Ignore small angles
    if abs(angle_now) <= 2:
        angle_now = 0
        servo_angle_z = 150
        Lastbit.set_servo('S2', servo_angle_z)
    else:
        # Calculate target angle with limits
        pitch_target = int(servo_angle_z - angle_now * KP)
        pitch_target = max(80, min(150, pitch_target)) # Limit
        Lastbit.set_servo('S2', pitch_target)

    # Get roll angle
    angle_row = Lastbit.get_roll()

    # Ignore small angles
    if abs(angle_row) <= 2:
        angle_row = 0
        Servo_angle = 90
        Lastbit.set_servo('S3', Servo_angle)
    else:
        # Calculate target angle with limits
        roll_target = int(Servo_angle + angle_row * (-KP_row))
        roll_target = max(20, min(150, roll_target)) # Limit
        Lastbit.set_servo('S3', roll_target)

    sleep(50)

```

Experimental Phenomenon

When you lift the front of the car, the car will control the vertical servo to move downward.



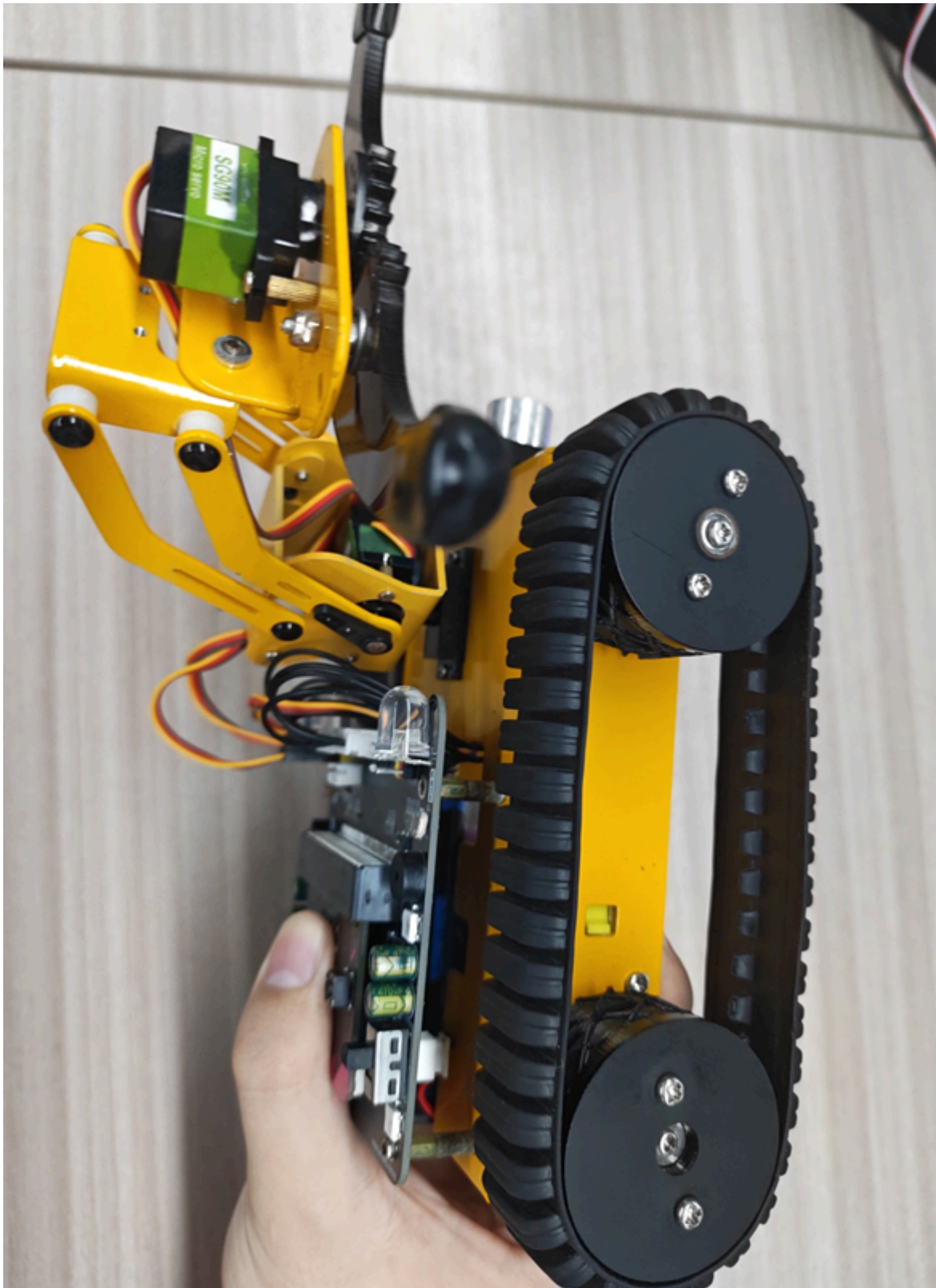
When you lift the rear of the car, the car will control the vertical servo to move upward.



When the car body tilts to the right, the car will control the horizontal servo to move to the left.



When the car body tilts to the left, the car will control the horizontal servo to move to the right.



Notes

- Make sure the servos are correctly connected and the battery is fully charged to drive the servo movement.
- When moving the car, it is recommended to move it slowly. Excessive rotation may cause jitter.